



MATHEMATICS

Multi-Digit Multiplication Warm-Up Week

Grade 5 | Warm-Ups

Build accurate, fluent standard-algorithm multiplication with estimation and error analysis.

INSIDE THIS RESOURCE

Teacher guidance • 16 practice prompts • Reflection exit ticket • Complete answer guidance

Print the practice pages separately from the teacher key. Provide extra paper for extended work.



Teacher guide

NC.5.NBT.5 - Demonstrate fluency with multiplication of whole numbers up to three digits by two digits using the standard algorithm.

Suggested time: Five 10-minute warm-ups

Learning focus: Build accurate, fluent standard-algorithm multiplication with estimation and error analysis.

Teaching sequence: Review prior instruction and model the first prompt with its answer. Assign one four-prompt page at a time. Ask for a check, a reason, or a specific example. Use the reflection ticket to plan the next lesson; it is not a standardized mastery score.

Five-day schedule: prompts 1–3, 4–6, 7–9, 10–12, then 13–16. Use separate paper for algorithms.

Model prompt: 34×26

Model answer: 884

Support options

AIG: justify a second method or create a transfer example. 504: follow the learner's plan for time, breaks, format, and reduced copying. EC: model steps and accept aligned oral, drawn, or written responses. ESL: preview vocabulary, use bilingual references, and rehearse before writing.

Source and scope: North Carolina Department of Public Instruction, Mathematics Standards Vertical Progression: <https://www.dpi.nc.gov/documents/publications/catalog/ma196-vertical-progression-math/open>; NC Check-Ins 2.0

Mathematics Specifications:

<https://www.dpi.nc.gov/documents/accountability/testing/nc-checkin/nc-check-ins-20-mathematics-grades-3-8-specifications> (accessed September 14, 2026).



Practice 1–4

Show work or explain your response. Use additional paper when needed. Teachers may cut apart the four bordered prompts.

1. 34×26

2. 128×14

3. 305×22

4. Estimate 247×31 using compatible numbers.



Practice 5–8

Show work or explain your response. Use additional paper when needed. Teachers may cut apart the four bordered prompts.

5. 46×19

6. 215×43

7. A printer makes 125 pages per minute for 24 minutes.

8. Check this answer: $63 \times 27 = 1,071$. Show correct partial products.



Practice 9–12

Show work or explain your response. Use additional paper when needed. Teachers may cut apart the four bordered prompts.

9. 408×16

10. 72×35

11. Which is reasonable for 296×18 : 532, 5,328, or 53,280?

12. A theater has 36 rows of 128 seats.



Practice 13–16

Show work or explain your response. Use additional paper when needed. Teachers may cut apart the four bordered prompts.

13. 504×29

14. 99×47

15. Use a second method to check 214×32 .

16. Explain why the second partial product shifts one place.



Reflection exit ticket

Complete independently. Refer to your completed practice, but explain in your own words. This checks reflection and application, not a test score prediction.

1. State the focus of this resource in your own words.
2. Create one accurate example connected to the practice.
3. Explain how you checked that example.
4. Identify one possible error and show a correction.
5. Name one specific next practice step and explain why it will help.



Teacher answer guidance

1. 884
2. 1,792
3. 6,710
4. About 7,500
5. 874
6. 9,245
7. 3,000 pages
8. $63 \times 20 = 1,260$; $63 \times 7 = 441$; total 1,701. The final answer alone does not show which step was mistaken.
9. 6,528
10. 2,520
11. 5,328
12. 4,608 seats
13. 14,616
14. 4,653
15. 6,848; e.g., $214 \times 30 + 214 \times 2$
16. It represents multiplication by tens, not ones.

Reflection criteria

Items 1–4: 2 points for an accurate, specific response; 1 for a partly correct response; 0 for a missing or unrelated response. Accept justified alternatives. Item 5 is an unscored planning reflection. Feedback should identify a next teaching move, not label a learner.

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